Satyam Kale

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PROFESSIONAL SUMMARY

Data Professional with nearly 2 years of experience in data engineering, analytics, and data science. Built production ETL pipelines on AWS and GCP, dimensional warehouse models in Redshift and Snowflake, Power BI and Tableau dashboards tracking 30+ KPIs, and ML models (XGBoost, Random Forest, ARIMA) for credit risk, fraud detection, and forecasting. Proficient in Python, SQL, PySpark, and Apache Airflow. M.S. Data Science, UMD (GPA 3.7).

EDUCATION

University of Maryland, College Park Aug 2024 - May 2026
M.S., *Data Science* GPA: 3.7/4.0

Savitribai Phule Pune University Aug 2020 - Jun 2024
B.E., *Computer Science*, Minor: Data Science GPA: 3.7/4.0

SKILLS

Programming Languages: Python, C++, SQL, Shell, Java

Frameworks & Libraries: PyTorch, TensorFlow, JAX, scikit-learn, Hugging Face Transformers, LLMs, Diffusion Models, Neural Networks, LangChain, LlamaIndex, Sentence Transformers, spaCy, NLTK, Flask, Streamlit

Databases: FAISS, Pinecone, Vector Databases, BigQuery

Tools & Software: ONNX, GPT-4, Claude, Prompt Engineering, Quantization (INT8, FP16, AWQ), Retrieval-Augmented Generation (RAG), Agentic Workflows, Context Engineering, Semantic Search, Embedding Pipelines, vLLM, TensorRT-LLM, SGLang, NVIDIA Triton Inference Server, FlashInfer, Continuous Batching, KV-Cache Management, Docker, Git, CI/CD, Apache Airflow, Pandas, NumPy, OCR (Tesseract), A/B Testing, Model Evaluation, Error Analysis, Observability, python-pptx, Whisper API

Cloud Platforms: AWS (S3, EC2, SageMaker), GCP (BigQuery, Cloud Storage, Composer), Power BI, Power Automate, HubSpot, Salesforce, Microsoft Excel, API Integrations, Data Analytics, Dashboard Development, Workflow Automation, CRM Customer Potential Modeling, JIRA, Business Intelligence Applications, Financial Reporting Automation, Marketing Workflow Automation

WORK EXPERIENCE

University of Maryland, College Park | Graduate Research Assistant May 2025 - May 2026

- Researched and benchmarked LLM inference systems on UMD's HPC cluster (Zaratan), evaluating TensorRT-LLM across multiple quantization strategies (FP8, INT4/AWQ/GPTQ, BF16) on NVIDIA A100 GPUs, measuring tokens-per-second, time-to-first-token (TTFT), and memory efficiency at scale, then translating raw metrics into clear, actionable insights for leadership.
- Deployed and evaluated NVIDIA Triton Inference Server with TensorRT-LLM backend, enabling dynamic batching and pre/post-processing pipelines, reducing serving overhead and improving multi-tenant throughput.
- Designed and ran systematic multi-GPU distributed inference experiments using Tensor Parallelism (TP) and Pipeline Parallelism (PP), scaling 70B+ parameter models across 4-8 GPU configurations to characterize latency/throughput tradeoffs.
- Evaluated continuous batching and PagedAttention scheduling under varying concurrency levels, measuring p50/p99 latency and KV-cache utilization to identify optimal serving configurations for production workloads.

AICTE IDEA Lab and Innovation Centre | ML Inference Engineer Aug 2023 - Jun 2024

- Built and optimized end-to-end LLM inference pipelines for multilingual production models, covering API design, model serving, quantization, and performance benchmarking across the full stack, including API integrations with downstream systems.

- Implemented quantization strategies (INT8, FP16) and model graph optimizations reducing memory footprint and enabling larger batch sizes, improving hardware utilization across production GPU fleets.
- Optimized KV-cache management and attention scheduling for high-throughput multi-tenant serving, improving sustained throughput under concurrent user loads.
- Integrated ML model changes into CI/CD pipelines, conducting systematic A/B benchmarks to validate latency and throughput across hardware configurations.
- Collaborated cross-functionally with model, platform, and product teams to translate research improvements into production-ready inference system upgrades, communicating recommendations clearly across multiple functions.

PROJECTS

CAFBrain – RAG Document Generation System

- Designed and built an end-to-end RAG pipeline automating content generation (grant proposals, reports, blog posts, presentations) from a large corpus of existing nonprofit documents (PDFs, DOCX, PPTs), eliminating repetitive manual drafting for CAFB staff.
- Built the full ingestion and retrieval stack: OCR-based document preprocessing (Tesseract, spaCy, NLTK), chunk-level embedding with Sentence Transformers, vector indexing in FAISS/Pinecone, and retrieval orchestration via LangChain/LlamaIndex.
- Integrated GPT-4 for context-aware draft generation with prompt engineering for tone and style control; implemented multi-modal output conversion to PowerPoint (python-pptx) and video scripts (Whisper API).
- Delivered a Streamlit human-in-the-loop interface enabling staff to review, edit, and refine AI-generated drafts before publication.

LLM Inference Pipeline – GPU Performance Optimization

- Built an end-to-end PyTorch-based LLM inference pipeline for semantic question answering, applying kernel optimizations and batched inference strategies that improved inference throughput by 20% under concurrent load.
- Deployed via Flask REST API with sub-100 ms response time; implemented GPU memory-hierarchy tuning to maximize throughput at scale.

Hospital Readmission Prediction – ML Pipeline

- Designed an ETL and feature engineering pipeline in Python and BigQuery processing 100K+ healthcare records; trained and evaluated logistic regression and random forest models, improving readmission prediction accuracy by 15% (ROC-AUC optimized).
- Orchestrated automated batch ML workflows with Apache Airflow, scheduling ingestion, feature generation, and model-ready dataset delivery.

OPEN-SOURCE CONTRIBUTIONS

NVIDIA/TensorRT-LLM

- PR #17478 Added pipeline diagram for the Overlap Scheduler to official TensorRT-LLM documentation, visualizing concurrent GPU/CPU execution across decoding steps.
- PR #17747 Implemented Google-style docstrings with type annotations for the CachedModelLoader class, docu menting model loading, MPI dispatch, and speculative draft model resolution behavior.

vllm-project/vLLM

- vLLM PR #52259 Added Disaggregated Batch Overlap (DBO) schedule diagram clarifying the overlap scheduling pattern for prefill and decode phases.
- dynamo PR #13043 Added Planner control loop architecture diagram to NVIDIA Dynamo inference framework documentation.